

Recall IPS: Inner Product space

$F = \mathbb{C}$ or $\mathbb{R} \rightsquigarrow$ geom $\rightsquigarrow$ metric distance approximation

Def $V \times V \rightarrow F$

$$(x, y) \mapsto (x, y)$$

1. positivity $(x, x) \geq 0$ and $(x, x) = 0 \Leftrightarrow x = 0$

2. conjugate symmetry: $(x, y) = \overline{(y, x)}$

3. Bilinear $(a_1 x_1 + a_2 x_2, y) = a_1 (x_1, y) + a_2 (x_2, y)$

• $(V, (\cdot, \cdot))$ — IPS/ F

Thm $\dim V = n < +\infty$, $(\cdot, \cdot)$ is an IP over V . For any basis $\mathcal{B} = \{v_1, \dots, v_n\}$ of V , there exists uniquely a positive matrix A s.t. $(u, v) = y^* A x$, where x resp y is the coordinate of u and v with $\mathcal{B}$

Def $\forall x, y \in F^n$ $(x, y) \stackrel{\text{def}}{=} y^* A x$ is an IP

• What's A is Thm

$$\begin{aligned} y^* A x &= (u, v) = \left(\sum_{i=1}^n x_i v_i, \sum_{j=1}^n y_j v_j \right) \\ &= \sum_{i,j} x_i \overline{y_j} (v_i, v_j) \\ &= \sum_{i,j} a_{ij} x_i \overline{y_j} = y^* A x \end{aligned}$$

where $A = (a_{ij}) = ((v_i, v_j))_{n \times n}$

★ How to define the angles?

Cauchy-Schwarz's Inequality $|(u, v)| \leq \|u\| \|v\|$ over $\mathbb{R}$

$$\text{so } -1 \leq \frac{(u, v)}{\|u\| \|v\|} \leq 1 \quad \theta = \arccos \frac{(u, v)}{\|u\| \|v\|}$$

Can define orthogonality 正交性

Recall. $Ax = b$ $\begin{cases} \text{consistent} & x = x_0 + N(A) \\ \text{inconsistent} & \hat{x} = (A^* A)^{-1} A^* b \end{cases}$

$$P(A_{m \times n} X = b) \begin{cases} 1 & m \leq n \\ 0 & m > n \end{cases}$$

Ex. Fourier Series

$$1. \quad V = \mathcal{F}[x], \quad (f, g) = \int_0^1 f \overline{g} dx$$

$$2. 1 \sin x \cos x \dots \sin nx \cos nx \dots$$

$$\int_{-\pi}^{\pi} \sin nx \cos mx dx = 0$$

$$\frac{1}{\pi} \int_{-\pi}^{\pi} \sin nx \sin mx dx = \begin{cases} 0 & m \neq n \\ 1 & m = n \end{cases}$$

问: x 轴, y 轴的夹角可以定义成什么?

$$\mathbb{R}^2: \omega(e_1, e_2) \neq \pm 1 \quad \langle e_1, e_2 \rangle \neq k\pi$$

• Comparing V and W (IPS)

$$\sigma: V \longrightarrow W \text{ s.t. } \sigma(v_1), \sigma(v_2) = (v_1, v_2)$$

$$\text{② } \|\sigma(v)\| = \|v\|$$

Def Let V, W be two IPS. $\sigma \in \text{Hom}(V, W)$ is an isometry if

$$\|\sigma(v)\| = \|v\|, \forall v \in V$$

observation:

1. An isometry is inj. so $\text{Im } \sigma \cong V$

so enough to study $\text{End } V$

so if $\dim V < \infty$, enough to study isom $\sigma \in \text{Aut } V$ (automorphism) 自同构.

2. Let $\sigma \in \text{Aut. } V$ be an isom. Then $(\sigma(u), \sigma(v)) = (u, v) \forall u, v \in V$

$$\text{iff. } (\sigma(u+v), \sigma(u+v)) = (u+v, u+v)$$

$$(\sigma(u) + \sigma(v), \sigma(u) + \sigma(v)) = (u, u) + (u, v) + (v, u) + (v, v)$$

$$\sigma(u)^2 + 2(\sigma(u), \sigma(v)) + \sigma(v)^2 =$$

$$(\sigma(u), \sigma(v))$$

Let the matrix of σ under β is M

$$\text{Then } \sigma(v) = (v_1, \dots, v_n) M$$

$$\text{so } (\sigma(u), \sigma(v)) = x^* M^* A M x = x^* A x$$

$$\text{so } M^* A M - A = 0$$

Conclusion Any finite dim IPS V admits ONB (orthogonal basis)

Thm Let $\sigma \in \text{Aut } V$ be an isometry. Then its matrix under an ONB

is unitary. ad hoc any isometry over ^(unitary space) Euclidean space is orthogonal transformation

$$(O_n / \mathbb{R}, U_n / \mathbb{C})$$

EX. R

①

$$n=1 \pm 1$$

$$e^{ix}$$

$$n=2 \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$$

$$\begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix}$$

$$n=3 \left\{ \begin{array}{l} |A(A)R| = 1 (\text{if } \pm 1) \\ |A(A)R| = 3 (\pm 1) \end{array} \right.$$

$$|A(A)R| = 3 (\pm 1)$$